Volatility Factor Analysis

Cross-Sectional Return Prediction Study

Portfolio Builder – Track A

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1 Objective

The objective of this analysis is to determine whether historical volatility can be used as a useful cross-sectional factor for predicting future stock returns.

The analysis follows the same research framework used for the momentum factor:

1. Construct a volatility signal for every stock.
2. Measure its relationship with future returns using cross-sectional Information Coefficient (IC).
3. Divide stocks into five volatility groups.
4. Compare the average future return of the groups.
5. Determine whether a systematic relationship exists.
6. Evaluate whether volatility is suitable as a return-ranking factor.

The analysis uses daily OHLCV data from the project dataset.

2 Volatility Definition

We use **20-day historical volatility** as the volatility signal.

First, daily returns are calculated as

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1, \tag{1}$$

where

- $P_{i,t}$ is the closing price of stock i on day t,
- $r_{i,t}$ is the daily return of stock i.

The 20-day historical volatility is then calculated as the standard deviation of the most recent 20 daily returns:

$$\sigma_{i,t}^{(20)} = \mathrm{StdDev}\left(r_{i,t-19}, r_{i,t-18}, \ldots, r_{i,t}\right). \tag{2}$$

Thus, each stock receives a volatility score at every point in time.

A larger value of $\sigma_{i,t}^{(20)}$ indicates that the stock has experienced larger fluctuations in its recent daily returns.

3 Forward Return

To determine whether volatility predicts future performance, we calculate the 20-day forward return:

$$F_{i,t}^{(20)} = \frac{P_{i,t+20}}{P_{i,t}} - 1. \tag{3}$$

Therefore, for each stock and date we have:

$$\boxed{\text{20-day volatility} \quad \longrightarrow \quad \text{20-day future return}}$$

The question being tested is whether stocks with different levels of historical volatility subsequently generate systematically different returns.

4 Cross-Sectional IC Analysis

For every trading day, the volatility of all stocks is compared with their subsequent 20-day returns.

The cross-sectional Information Coefficient is defined as

$$IC_t = \mathrm{Corr}_i \left(\sigma_{i,t}^{(20)}, F_{i,t}^{(20)} \right). \tag{4}$$

The mean IC is then

$$\overline{IC} = \frac{1}{T} \sum_{t=1}^{T} IC_t. \tag{5}$$

The resulting statistics were:

Table 1: 20-Day Volatility vs. 20-Day Forward Return

Statistic	**Value**
Mean IC	−0.003914
Median IC	−0.013230
IC Standard Deviation	0.260588
Positive IC Fraction	48.27%

5 Interpretation of the IC Results

The mean IC is

$$\boxed{\overline{IC} = -0.003914}$$

which is extremely close to zero.

This indicates that, on average, there is almost no cross-sectional relationship between 20-day historical volatility and subsequent 20-day returns.

The median IC is also negative:

$$\boxed{\text{Median}(IC) = -0.01323}.$$

However, its magnitude is still very small.

Furthermore, only

$$\boxed{48.27\%}$$

of the daily IC values are positive.

If volatility had a stable positive relationship with future returns, we would expect positive ICs to occur substantially more often than 50%. Similarly, a strong inverse relationship would produce predominantly negative ICs.

Instead, the positive IC fraction is close to 50%, suggesting that the relationship changes considerably from one period to another.

Therefore, the IC analysis does not provide evidence of a stable cross-sectional return-ranking signal.

6 Pooled Correlation

A pooled correlation was also calculated by combining all stock-date observations:

$$\text{Corr}\left(\sigma_{i,t}^{(20)}, F_{i,t}^{(20)}\right) = 0.0672. \tag{6}$$

Thus,

$$\boxed{\rho_{\text{pooled}} = 0.0672}.$$

Although this correlation is positive, its magnitude is very small.

More importantly, pooled correlation should not be confused with the cross-sectional IC.

The two quantities answer different questions.

The pooled correlation considers all observations together:

$$\text{Corr}\left(\sigma_{i,t}^{(20)}, F_{i,t}^{(20)}\right).$$

The IC first calculates a correlation across stocks for each date:

$$IC_t = \mathrm{Corr}_i\left(\sigma_{i,t}^{(20)}, F_{i,t}^{(20)}\right),$$

and then averages those correlations:

$$\overline{IC} = \frac{1}{T}\sum_t IC_t.$$

For a cross-sectional factor strategy, the second measure is more relevant because the strategy ranks stocks relative to one another at each rebalancing date.

The important result is therefore

$$\boxed{\overline{IC} \approx 0}.$$

7 Five-Group Portfolio Test

To make the relationship easier to interpret, stocks were sorted into five groups according to their 20-day historical volatility.

- Group 1: Lowest volatility
- Group 2: Low-to-moderate volatility
- Group 3: Medium volatility
- Group 4: Moderate-to-high volatility
- Group 5: Highest volatility

The average subsequent 20-day return for each group was calculated.

Table 2: Average 20-Day Forward Return by Volatility Group

Volatility Group	**Average Forward Return**
Group 1 (Lowest)	1.7943%
Group 2	0.7115%
Group 3	1.4532%
Group 4	1.4390%
Group 5 (Highest)	3.0030%

8 Interpretation of the Group Results

The group returns are

$$1.7943\%, \quad 0.7115\%, \quad 1.4532\%, \quad 1.4390\%, \quad 3.0030\%.$$

There is no monotonic relationship between volatility and future return.

For a useful return-ranking factor, we would ideally observe a systematic pattern such as

$$R_1 > R_2 > R_3 > R_4 > R_5$$

or

$$R_1 < R_2 < R_3 < R_4 < R_5.$$

Instead, the observed sequence is

$$\boxed{1.7943\% \to 0.7115\% \to 1.4532\% \to 1.4390\% \to 3.0030\%}$$

which is irregular.

In particular, the highest-volatility group has the largest average forward return, but the intermediate groups do not follow a consistent pattern.

Therefore, simply increasing or decreasing exposure according to volatility would not produce a stable return-ranking rule based on these results alone.

9 Why Volatility Is Not a Good Return Indicator

The evidence from the analysis suggests that 20-day historical volatility is not a strong standalone indicator for predicting cross-sectional future returns.

There are several reasons.

9.1 1. Mean IC is Almost Zero

The strongest evidence is

$$\boxed{\overline{IC} = -0.003914}.$$

An IC close to zero means that the ranking produced by volatility has very little consistent relationship with the subsequent return ranking.

Thus, if two stocks have different historical volatility, their volatility levels provide little information about which one will have the higher 20-day future return.

9.2 2. IC Changes Sign Frequently

Only

$$48.27\%$$

of daily IC observations are positive.

Therefore, the relationship is not consistently positive or negative.

A useful cross-sectional signal should generally maintain a reasonably stable direction across time.

Here, the evidence instead suggests that the relationship is unstable.

9.3 3. No Monotonic Group Pattern

The five-group test gives

$$1.7943\%, \quad 0.7115\%, \quad 1.4532\%, \quad 1.4390\%, \quad 3.0030\%.$$

The lack of monotonicity means that moving from low volatility to high volatility does not correspond to a predictable change in future returns.

9.4 4. Pooled Correlation Is Also Weak

The pooled correlation is

$$\rho = 0.0672.$$

Although positive, this is small and does not establish a useful cross-sectional ranking relationship.

More importantly, the cross-sectional IC analysis gives a value very close to zero.

10 Important Distinction: Prediction vs. Risk Management

The conclusion that volatility is not a useful return-ranking factor does **not** mean that volatility is useless in portfolio construction.

Volatility can serve an entirely different purpose.

It can be used as a measure of risk when determining position sizes.

For example, a simple inverse-volatility weighting scheme can be written as

$$w_i \propto \frac{1}{\sigma_i}. \tag{7}$$

After normalization,

$$w_i = \frac{1/\sigma_i}{\sum_{j=1}^{N} 1/\sigma_j}. \tag{8}$$

Under such a framework, stocks with higher estimated volatility receive smaller positions, while stocks with lower estimated volatility receive larger positions.

This is fundamentally different from saying that volatility predicts returns.

Therefore:

Volatility may be useful for sizing even if it is not useful for ranking.

This distinction is particularly important for Track A, where the strategy is required to use volatility-adjusted position sizing.

11 Comparison with Momentum Analysis

The volatility factor behaved differently from the momentum factor studied previously.

For momentum, the cross-sectional analysis produced a consistently negative relationship between past momentum and future returns, including a stronger negative relationship for the 20-day lookback and 20-day forward horizon.

For volatility, however,

$$\overline{IC} = -0.003914$$

is approximately zero, and the five volatility groups do not exhibit a systematic progression in future returns.

Thus, the two factors provide different types of information:

Factor	**Return Ranking Evidence**	**Potential Role**
Momentum	Evidence of reversal relationship	Return signal
Volatility	Little evidence of return ranking	Risk / position sizing

12 Limitations

The conclusion should be interpreted within the limits of the current analysis.

1. Only a 20-day historical volatility specification has been tested in this section.

2. The relationship may depend on the volatility estimation window.

3. The five-group analysis is descriptive and does not by itself establish statistical significance.

4. Transaction costs, turnover, market impact and other implementation effects have not been incorporated into this factor test.

5. The results should ultimately be evaluated on the required out-of-sample dataset.

Therefore, the appropriate conclusion is not that volatility can never predict returns, but rather that **the tested 20-day historical volatility measure does not provide sufficient evidence of a useful cross-sectional return-ranking signal in this dataset**.

13 Final Conclusion

The 20-day historical volatility factor was evaluated using both cross-sectional IC analysis and a five-group portfolio test.

The main results were:

$$\boxed{\overline{IC} = -0.003914}$$

$$\boxed{\text{Median}(IC) = -0.013230}$$

$$\boxed{\text{Positive IC fraction} = 48.27\%}$$

and the average 20-day forward returns across the five volatility groups were

$$\boxed{1.7943\%,\ 0.7115\%,\ 1.4532\%,\ 1.4390\%,\ 3.0030\%}$$

The IC is very close to zero, the sign of the relationship is not stable, and the group returns do not form a monotonic pattern.

Therefore, based on the evidence obtained from this dataset,

$$\boxed{\text{20-day historical volatility is not a strong standalone cross-sectional return-ranking factor.}}$$

However, volatility remains relevant for the overall Track A strategy because it can be used for **risk estimation and volatility-adjusted position sizing**.

The research therefore separates the two roles:

$$\boxed{\text{Return signal} \neq \text{Risk measure}}$$

Volatility does not appear sufficiently informative as the former in this analysis, but it remains potentially useful as the latter.